Università degli Studi di Napoli
Federico II
Dipartimento di Ingegneria Elettrica e delle Tecnologie dell'Informazione

Statistical Learning and Data Analysis – Module A

TWO WAY CROSS CLASSIFICATION

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What further insights about the data?

- Exploratory Data Analysis aims to to convert *Data into Information.*
- **Two-way Cross-Classification** of data provide insights about the **bivariate or bidimensional frequency distribution** and the *unidimensional frequency distributions of the two single variables* as well.
- The aim is to study the relationship between a couple of variables in terms of association, correlation and linear correlation.
- In Data Mining and Machine Learning Module A Course you will learn about Multidimensional Data Analysis.

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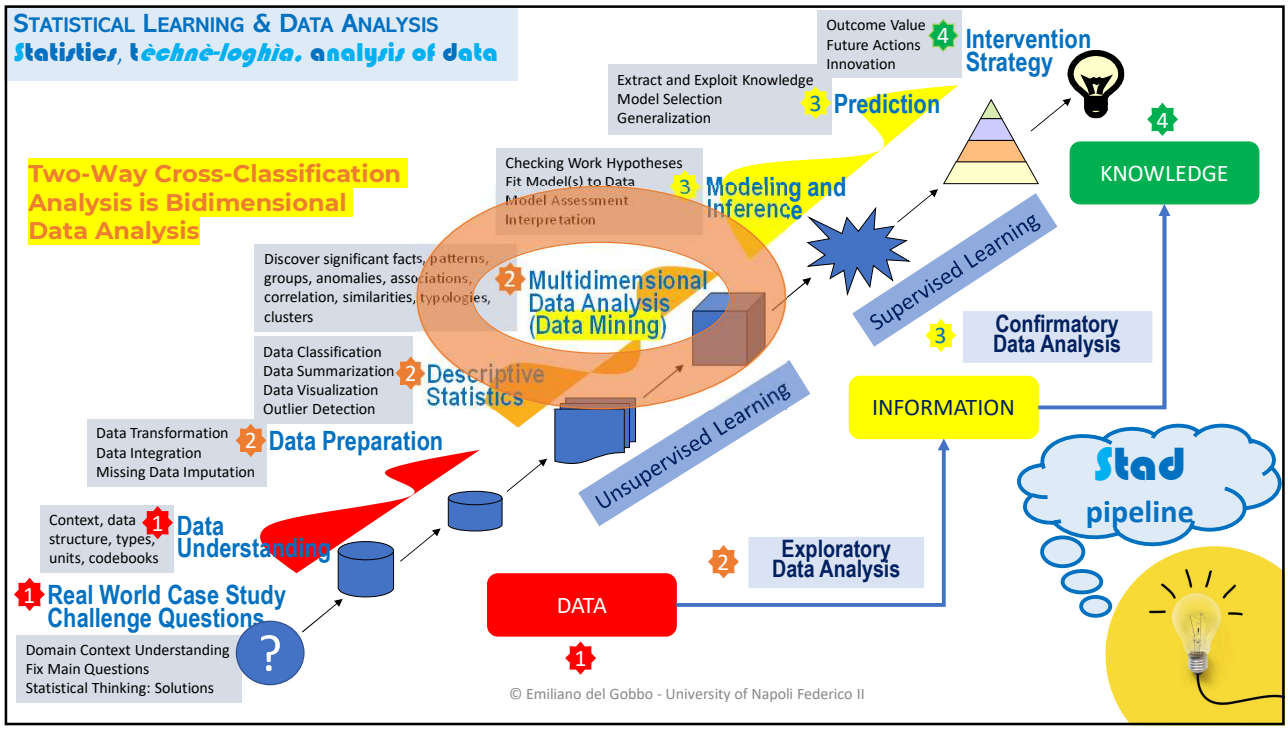

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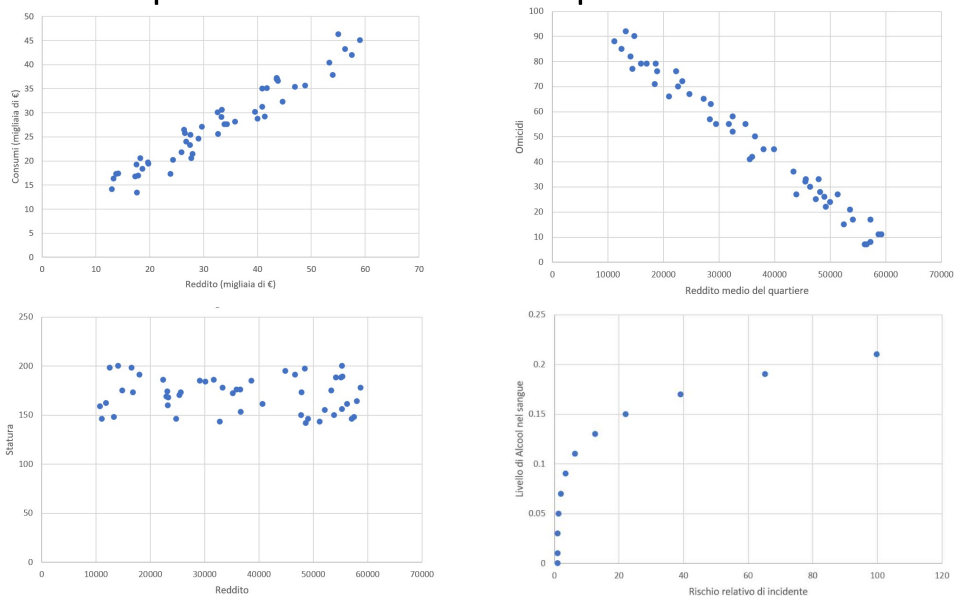
Multivariate quantitave scatter-plot



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Cross-Classification Table

- Two-way cross-classification table is a bivariate frequency distribution of two variables (X, Y).
- It consists in cross-classifying the units with respect to the distinct modalities of the row variable X as well as of the column variable Y : $\{(x_i, y_j), n_{ij}\}$ for $i = 1, \dots, I$ and $j = 1, \dots, J$ such that $\sum_{i=1}^{I}\sum_{j=1}^{J} n_{ij} = n$ where
- $\sum_{i=1}^{I} n_{ij} = n_{+j}$, for $j = 1, \dots, J$ are column margins with $\sum_{j=1}^{J} n_{+j} = n$
- $\sum_{j=1}^{J} n_{ij} = n_{i+}$, for $i = 1, \dots, I$ are row margins with $\sum_{i=1}^{I} n_{i+} = n$

Y / X	y_1	…	y_j	…	y_J	
x_1	n_{11}	…	n_{1j}	…	n_{1J}	n_{1+}
…	…	…	…	…	…	…
x_i	n_{i1}	…	n_{ij}	…	n_{iJ}	n_{i+}
…	…	…	…	…		…
x_I	n_{I1}	…	n_{Ij}	…	n_{IJ}	n_{I+}
	n_{+1}	…	n_{+j}	…	n_{+J}	n

The rows and columns represent two different stratifications of a sample.

✓*How many units are both male and use glasses?*

✓*Are more Male or Female that use glasses?*

✓*What about Male and Female not using glasses?*

✓*What is the proportion of males using glasses? What the proportion of females using glasses? Which of the two is higher?*

Use Glasses? Gender	NO	YES	*Total*
Male	20	40	*60*
Female	20	20	*40*
Total	*40*	*60*	**100**

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Marginal and Partial Frequency Distributions

- Two-way cross-classification includes not only the bidimensional frequency distribution (X, Y) but also **two Marginal Frequency Distributions** of the two single variables $X$and $Y$and $I + J$ **Partial or Conditional Frequency Distributions** $Y|X = x_i$ «Y given that $X = x_i$» and $X|Y = y_j$ «X given that $Y = y_j$».
- **Column and Row Marginal Frequency Distributions:** (y_j, n_{+j}) for $j = 1, \dots, J$, and (x_i, n_{i+}) for $i = 1, \dots, I$.
- ***I Partial or Conditional Frequency Distributions*** $Y|X = x_i$: (y_j, n_{ij}) or $(y_j, \frac{n_{ij}}{n_{i+}})$ for $j = 1, \dots, J$
- ***J Partial or Conditional Frequency Distributions*** $X|Y = y_j$: (x_i, n_{ij}) $(x_i, \frac{n_{ij}}{n_{+j}})$ or for $i = 1, \dots, I$.

Y / X	y_1	…	y_j	…	y_J	
x_1	n_{11}	…	n_{1j}	…	n_{1J}	n_{1+}
…	…	…	…	…	…	…
x_i	n_{i1}	…	n_{ij}	…	n_{iJ}	n_{i+}
…	…	…	…	…		…
x_I	n_{I1}	…	n_{Ij}	…	n_{IJ}	n_{I+}
	n_{+1}	…	n_{+j}	…	n_{+J}	n

✓*40/60 of the males use glasses while 20/40 of the females use glasses and 60/100 of the overall sample use glasses.*

✓*This table provides some insights that are more males than females in this sample that use glasses!*

Use Glasses? Gender	NO	YES	*Total*
Male	20	40	*60*
Female	20	20	*40*
Total	*40*	*60*	**100**

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Two-Way Cross-Classification Data Analysis

Categorical Data Analysis or Contingency Table Analysis

- Both variables are nominal
- Aim is to analyse the gap from **independence in distribution**
  - **Association**: Symmetric or Interdependence Data Analysis
  - **Predictability**: Non-Symmetric or Dependence Data Analysis

Correlation Analysis

- One variable is numerical playing the role of target or response variable and the other can be either nominal or numerical playing the role of stratifying factor
- Aim is to analyse the gap from **independence in mean**
  - **Correlation** but in Non-Symmetric way for Dependence Data Analysis

Linear Correlation Analysis

- Both variables are numerical
- Aim is to analyse the **linear correlation**

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Contingency Table Analysis: Row and Column Profiles

- Let consider the relative frequency distribution $f_{ij} = \frac{n_{ij}}{n}$ with $\sum_{i=1}^{k}\sum_{j=1}^{h} f_{ij} = 1$.
- The generic j-th Conditional Frequency Distribution $X|Y = y_j$ provides the j-th **column profile**. It holds: $\sum_{i=1}^{k} \frac{f_{ij}}{f_{i+}} f_{i+} = f_{+j}$ for $j = 1, \dots, J$ **(centroid of the row profiles)**
- The generic i-th Conditional Frequency Distribution $Y|X = x_i$ provides the i-th **row profile.** It holds: $\sum_{j=1}^{h} \frac{f_{ij}}{f_{+j}} f_{+j} = f_{i+}$ for $i = 1, \dots, I$ **(centroid of the column profiles)**

$X \backslash Y$	y_1		y_j	...	y_J	
x_1	$\frac{f_{11}}{f_{1+}}$		$\frac{f_{1j}}{f_{1+}}$	...	$\frac{f_{1J}}{f_{1+}}$	1
...						
x_i	$\frac{f_{i1}}{f_{i+}}$		$\frac{f_{ij}}{f_{i+}}$	...	$\frac{f_{iJ}}{f_{i+}}$	1
...	...		...	...	...	...
x_I	$\frac{f_{I1}}{f_{I+}}$		$\frac{f_{Ij}}{f_{I+}}$	...	$\frac{f_{Ih}}{f_{I+}}$	1
	f_{+1}		f_{+j}	...	f_{+J}	

i-th ***Row profile***

Centroid of the row profiles

Averaging the I i-th elements weighted by f_{i+}

j-th element of the centroid

j–th ***Column profile***

$X \backslash Y$	y_1	...	y_j	...	y_J	
x_1	$\frac{f_{11}}{f_{+1}}$	...	$\frac{f_{1j}}{f_{+j}}$	...	$\frac{f_{1J}}{f_{+J}}$	f_{1+}
...						...
x_i	$\frac{f_{i1}}{f_{+1}}$	...	$\frac{f_{ij}}{f_{+j}}$	...	$\frac{f_{iJ}}{f_{+J}}$	f_{i+}
...	...	...	...	...	...	
x_I	$\frac{f_{I1}}{f_{+1}}$	...	$\frac{f_{Ij}}{f_{+j}}$	...	$\frac{f_{IJ}}{f_{+J}}$	f_{I+}
	1	...	1	...	1	

Averaging the J i-th elements weighted by f_{+j}

i-th element of the centroid

Centroid of the column profiles

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The independence in distribution

- $\frac{f_{ij}}{f_{i+}} = f_{+j}$
- $\frac{f_{ij}}{f_{+j}} = f_{i+}$
- $f_{ij} = f_{i+} f_{+j}$
- For any cell (i, j).
- All row profiles are equivalent to their centroid.
- All column profiles are equivalent to their centroid.
- Cross-classification is useless if there is independence in distribution.

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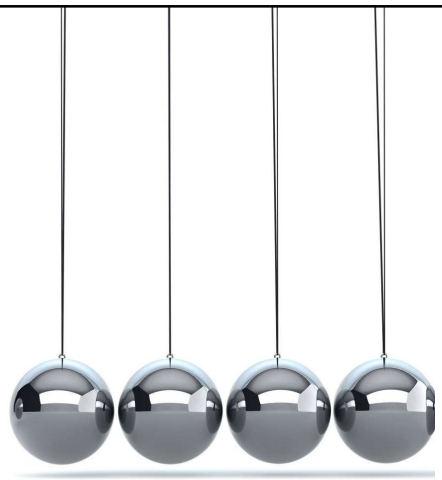

Association (Symmetric) Analysis

Pearson's Mean Square Contingency Coefficient

- The inertia of the (row and column) points with respect to the centroid or center of gravity is given by:

$$\Im = \sum_{i=1}^{k} \sum_{j=1}^{h} \frac{\left(f_{ij} - f_{i+} f_{+j}\right)^2}{f_{i+} f_{+j}} = \frac{\chi^2}{n} = \phi^2$$

- χ^2 is the chi-square index of association;
- $0 \leq \phi^2 \leq min(k-1, h-1)$.
- It is a symmetric coefficient since rows and columns play the same role in the analysis.

In machine learning terminology, it is known as Matthews correlation coefficient (MCC) so coined by the biochemist Matthews.

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Goodman and Kruskal τ index of predictability

- The predictability τ index of Goodman and Kruskal is the Non-Symmetric version of Pearson's χ^2 contingency coefficient for two-way cross-classifications where the two variables play a different role in the analysis such to predict the response categories on the basis of the predictor's categories.
- The τ index can be defined, for $i = 1, \dots, I$ predictor categories and $j = 1, \dots, J$ response categories, as:

Gini Heterogeneity Index

$$\tau_{Y|X} = \frac{\sum_j \sum_i \left(\frac{f_{ij}}{f_{i+}} - f_{+j}\right)^2 p_{i+}}{1 - \sum_j f_{+j}^2} = \frac{\sum_i \sum_j \frac{f_{ij}^2}{f_{i+}} - \sum_j f_{+j}^2}{1 - \sum_j f_{+j}^2} \qquad 0 \le \tau_{Y|X} \le 1$$

- where f_{ij} is the relative frequency, $\sum_j f_{ij} = f_{i+}$, $\sum_i f_{ij} = f_{+j}$, $\sum_i \sum_j f_{ij} = 1$, $\sum_j f_{i+} = 1$, $\sum_i f_{+j} = 1$.
- $\tau_{Y|X} = 1$ if for any i it holds: $f_{ij'} = f_{i+}$ and $f_{ij} = 0$ for $j \neq j'$ (maximum homogeneity of the response variable in any row).
- $\tau_{Y|X} = 0$ if for any i it holds: $\frac{f_{ij}}{f_{i+}} = f_{+j}$ (independence in distribution).

Dependence (Non-Symmetric) Analysis

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Examples

DISEASE GROUP	NO	YES	*Total*
A	20	40	*60*
B	30	60	*90*
C	20	40	*60*
Total	*70*	*140*	**210**

DISEASE GROUP	NO	YES	*Total*
A	60	0	*60*
B	0	90	*90*
C	60	0	*60*
Total	*120*	*90*	**210**

LEVEL OF RISK GROUP	LOW	MEDIUM	HIGH	*Total*
A	60	0	0	*60*
B	0	90	0	*90*
C	0	0	60	*60*
Total	*60*	*90*	*60*	**210**

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The independence in mean

- Say X is the row variable (of any type) playing the role of stratifying variable and Y is the column variable **of numerical type** playing the role of target variable:

$$\mu_{Y|X=x_i} = \mu_Y \text{ for any row } i = 1, \dots, I$$

where

- $\mu_{Y|X=x_i} = \sum_{j=1}^{h} y_j \frac{f_{ij}}{f_{i+}}$
- $\mu_Y = \sum_{j=1}^{h} y_j f_{+j}$

- The partial means $\mu_{Y|X=x_i}$ of the Y target variable for any distinct modality of the X stratifying variable are all equal to the general mean μ_Y of Y. This means that there is no difference in mean of the Y target variable when changing the modality of the stratifying factor.
- As an example: is there any difference in the mean Salary between Males and Females?
- **Property:** If there is independence in distribution there is also independence in mean, the *viceversa* is not true necessarily. You can try to demonstrate mathematically.

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Pearson's correlation coefficient

- The two variables play a different role in the analysis: there is a ***target or response variable*** Y to be explained and a ***stratifying factor or explanatory variable*** X that may have some effect on the target.
- The total Deviance of the target variable Y can be decomposed as follows:

Dependence (Non-Symmetric) Analysis

$$Dev(Y) = Dev(B) + Dev(W)$$

where:

- $Dev(B) = \sum_{i=1}^{k} \left(\mu_{Y|X=x_i} - \mu_Y\right)^2 f_{i+}$ is the **Between-Group Deviation**
- $Dev(W) = \sum_{i=1}^{k} \sum_{j=1}^{h} \left(y_j - \mu_{Y|X=x_i}\right)^2 f_{ij}$ is the **Within-Group Deviation**

- **Pearson's Correlation Coefficient** is defined as:

$$\eta^2_{Y|X} = \frac{Dev(B)}{Dev(Y)} = 1 - \frac{Dev(W)}{Dev(Y)}. \qquad 0 \le \eta^2_{Y|X} \le 1$$

- It is a *non-symmetric coefficient*. If both are numerical variables $\eta^2_{Y|X} \neq \eta^2_{X|Y}$.

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Bravais-Pearson Linear Correlation Coefficient

- Say X and Y are row and column variables, both numerical.
- Let $f_{ij} = \frac{n_{ij}}{n}$ be the relative frequency such that $\sum_{i=1}^{k}\sum_{j=1}^{h} n_{ij} = n$ and $\sum_{i=1}^{k}\sum_{j=1}^{h} f_{ij} = 1$.
- Covariance:

$$\sigma_{XY} = \sum_{i=1}^{k}\sum_{j=1}^{h}(x_i - \mu_X)(y_i - \mu_Y)f_{ij}$$

Dependence (Non-Symmetric) Analysis

- Variances:

$\sigma_X^2 = \sum_{i=1}^{k}(x_i - \mu_X)^2 f_{i+}$. $\quad \sigma_X^2 = \sum_{j=1}^{h}(y_j - \mu_Y)^2 f_{+j}$

- Cauchy-Schwarz inequality:

$$\sigma_{XY}^2 \leq \sigma_X^2\sigma_Y^2$$

- **Linear correlation coefficient** is a *symmetric* coefficient:

$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X\sigma_Y} \quad -1 \leq \rho_{XY} \leq +1$.

Correlation equal to -1 indicates a perfect negative correlation, 1 indicates a perfect positive correlation, and 0 indicates no linear correlation. The closer the value is to -1 or 1, the stronger the correlation. Values close to 0 suggest weak or no linear relationship.

- For both numerical variables, it can be shown that: $0 \leq \rho_{XY}^2 \leq \eta_{Y|X}^2 \leq +1$. *What does it mean?*

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Variance Decomposition

Number of units sold by product position on the shelf in 12 stores

Position on the shelf		
Top	Middle	Bottom
8	4	6
7	9	9
5	1	3
3	2	4

Exercise

- Check if sales are determined by the product's position on the shelf and write a short comment on the result found

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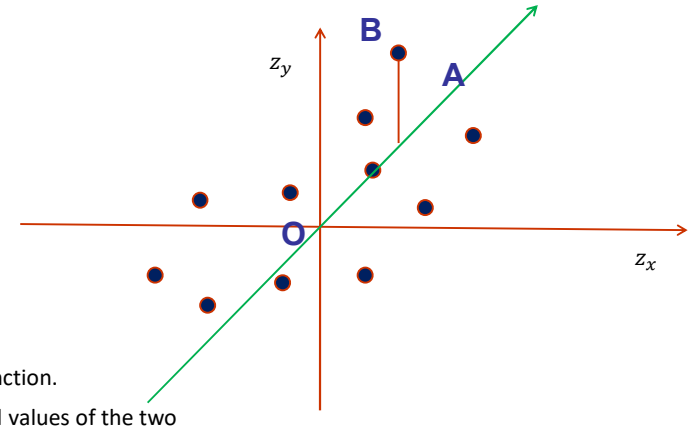
Another view of the linear coefficient



- Statistical Interpolation using a linear function.
- The mean of the product of standardized values of the two variables is equal to the Linear Correlation Coefficient: $\rho_{XY} = \frac{1}{n}\sum_{l=1}^{n} z_{x_l} z_{y_l}$ where $z_{x_l} = \frac{x_l - \mu_X}{\sigma_X}$ and $z_{y_l} = \frac{y_l - \mu_Y}{\sigma_Y}$

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Hierarchy and General Rules

- **Hierarchy** in Two-Way Cross-Classification Data Analysis:
  - Contingency Table Analysis can be also performed when one or both variables are numerical.
  - Correlation Analysis can be performed also when both variables are numerical.
  - Linear Correlation Analysis can be performed only when both variables are numerical.
- **General Rules**:
  - Independence in distribution implies independence in mean which implies no linear correlation.
  - The Vice-versa do not occur in any of the two rules.
- How can we use these rules?

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